



UNIVERSITY OF JOHANNESBURG

FACULTY OF SCIENCE

COMPUTER SCIENCE 1A

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Problem Description

Movement:

- The player may move one step in any direction, except diagonally.
- The player may not move outside the game world.
- The player may move to any location in the game world area if an immovable obstacle does not occupy it.
- With every turn (movement or attempted movement into an immovable object), the player loses 1 unit of energy.
- If the player moves over an energy pickup, the player gains 5 units of energy.
- If the player moves over a trap, the player loses 3 additional units of energy.
- If the player moves onto the goal location, the game ends in victory.

Input & Output

Input

<i>Input Description</i>	<i>Mechanism</i>
Movement choice(w,a,s,d,q)	Input Stream

Output

<i>Output Description</i>	<i>Stream (optional)</i>
Game grid	Output Stream

Data Format

<i>Identifier</i>	<i>Data Type</i>	<i>Description</i>
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rows	int	Stores and keeps track of position in 2D array row
cols	int	Stores and keeps track of position in 2D array column
Option	Char	Allows and stores movement option from user
twodarray	Int**	Pointer pointer of the 2D dynamic array
onedarray	Int*	Pointer of the 2D dynamic array

PSUEDOCODE

PROCEDURE MovePlayer(array, cols, rows, energyLevel, option)

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IF energyLevel <= 0 THEN
    PRINT "Defeat"
    SET continueGame TO false
    RETURN
ENDIF

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// Find current player position
CALL FindPlayer(array, rows, cols, PRow, PCol)

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SET Drow TO PRow
SET Dcol TO PCol

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// Determine movement direction
SWITCH UPPERCASE(option)
CASE 'W':
    Drow = Drow - 1 // move up
CASE 'A':
    Dcol = Dcol - 1 // move left
CASE 'S':

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        Drow = Drow + 1 // move down
CASE 'D':
    Dcol = Dcol + 1 // move right
ENDSWITCH

// Check if new position is inside bounds
IF IsInBounds(rows, cols, Dcol, Drow) THEN

    IF array[Drow][Dcol] == SPACE THEN
        MOVE player to new position
        SET old position to SPACE
        energyLevel = energyLevel - 1

    ELSE IF array[Drow][Dcol] == ENERGY THEN
        MOVE player to new position
        SET old position to SPACE
        energyLevel = energyLevel + 5
        PRINT "More Energy More Moves"

    ELSE IF array[Drow][Dcol] == OBSTACLE THEN
        PRINT "Player can't go this way"
        // No movement

    ELSE IF array[Drow][Dcol] == TRAP THEN
        MOVE player to new position
        SET old position to SPACE
        energyLevel = energyLevel - 3

    ELSE IF array[Drow][Dcol] == GOAL THEN
        MOVE player to new position
        SET old position to SPACE
        PRINT "Victory!"
        SET gameWin TO true
    ENDIF

ELSE
    PRINT "Move out of bounds!"
ENDIF

END PROCEDURE

```

UML Activity Diagram



